

Implementation Details

PINNs for 1D heat diffusion — repository `pinn-heat-diffusion_1D`

Quentin Pontalier

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1 Scope

This document describes, exhaustively, what the code in this repository does: every modelling choice, hyperparameter, random seed and evaluation protocol behind the figures and numbers in the `README`. A reader should be able to reimplement the results from this document alone. The scientific background is summarized in the technical note (`docs/note`).

2 Problem and synthetic data

2.1 Governing equations

$$\partial_t T = \alpha \partial_{xx} T, \quad x \in [0, L], \quad t \in [0, t_{\max}], \quad T(x, 0) = \sin(\pi x / L), \quad T(0, t) = T(L, t) = 0, \quad (1)$$

with $L = 1$ and $t_{\max} = 1.5$. The exact solution and its sensitivity to α (`pinn_heat/physics.py`) are

$$T(x, t) = \sin(\pi x/L) e^{-\alpha k^2 t}, \quad S_\alpha(x, t) = \frac{\partial T}{\partial \alpha} = -k^2 t T(x, t), \quad k = \pi/L. \quad (2)$$

2.2 Noisy observations

`data.noisy_observations(n, alpha, noise_level, seed)` proceeds as follows:

1. It creates a NumPy generator `np.random.default_rng(seed)`.
2. It draws n locations independently: $t_i \sim \mathcal{U}(0, t_{\max})$, then $x_i \sim \mathcal{U}(0, L)$.
3. It computes T_i^{clean} from the exact solution and sets $\sigma = \eta \cdot \text{std}(T^{\text{clean}})$, where η is `noise_level` and the standard deviation is taken over the n drawn points.
4. It adds noise: $T_i^{\text{obs}} = T_i^{\text{clean}} + \sigma \xi_i$, with $\xi_i \sim \mathcal{N}(0, 1)$.
5. It discards every observation with $T_i^{\text{obs}} < 0$ (`drop_negative=True`). Such values lie below the noise floor and contradict the physics of this problem. The number of points actually kept is reported (e.g. 24 of 25).

The noise level is therefore relative to the *spread* of the clean signal, not to each point's value. The exact solution is used only to generate data and to evaluate errors; it never enters the PINN loss.

2.3 Collocation points

The PDE residual is evaluated at $N_f = 500$ points drawn by Latin Hypercube Sampling: `scipy.stats.qmc.LatinHypercube(seed=seed)`, scaled to $[0, t_{\max}] \times [0, L]$. They are drawn once and kept *fixed* throughout training. This is required for L-BFGS, which builds its curvature model on a fixed objective.

3 Network

3.1 Architecture and initialization

`model.init_mlp(key, width=16, depth=3)` builds a fully connected network with layer sizes $[2, 16, 16, 16, 1]$. The input is the raw pair (t, x) in that order, with no normalization. Hidden layers use tanh; the output layer is linear. Each layer is initialized with Glorot-uniform weights and zero biases:

$$W \sim \mathcal{U}\left(-\sqrt{6/(n_{\text{in}} + n_{\text{out}})}, \sqrt{6/(n_{\text{in}} + n_{\text{out}})}\right), \quad b = 0. \quad (3)$$

The PRNG key `jax.random.PRNGKey(seed)` is split into one sub-key per layer. The network has 609 trainable weights and biases.

3.2 Hard-constrained ansatz

The temperature is not the raw network output:

$$T_\theta(x, t) = \sin(\pi x/L) + t x (L - x) \mathcal{N}_\theta(t, x). \quad (4)$$

The prefactor $t x (L - x)$ vanishes at $t = 0$, $x = 0$ and $x = L$. The initial and both boundary conditions therefore hold exactly for any weights, and no IC/BC loss terms are needed.

```
def temperature(layers, t, x, L=1.0):
    n = mlp(layers, jnp.stack([t, x]))[0]
    return jnp.sin(jnp.pi * x / L) + t * x * (L - x) * n
```

3.3 Diffusivity parameterization (inverse problem)

α is never trained directly. The trainable variable is β , and three mappings are implemented (`alpha_param`):

<code>alpha_param</code>	$\alpha(\beta)$	$d\alpha/d\beta$	initialization β_0
"square" (default)	$\alpha_{\min} + \beta^2$, $\alpha_{\min} = 10^{-12}$	2β	$\sqrt{\alpha_0 - \alpha_{\min}}$
"softplus"	$\log(1 + e^\beta)$	$1/(1 + e^{-\beta})$	$\log(e^{\alpha_0} - 1)$
"linear"	β	1	α_0

The default is $\alpha_0 = 1.0$. In the forward problem α is a fixed constant and only the network is trained.

Collapse to the trivial solution. The choice of mapping matters. With "softplus", the very first L-BFGS step moves β to a large negative value. On the datasets of Section 8 this gives $\alpha < 10^{-30}$ after one iteration. For $\beta \ll 0$, softplus saturates: $\alpha \approx e^\beta$ and $d\alpha/d\beta \approx \alpha \approx 0$. The gradient with respect to β therefore vanishes and α never recovers. The network then settles on a compromise around the trivial value $\alpha = 0$, where the PDE reduces to $\partial_t T = 0$; both loss terms stay between 10^{-2} and 10^{-1} .

With "square", $d\alpha/d\beta = 2\beta$ shrinks only linearly as $\alpha \rightarrow 0$ and never saturates exponentially. α dips to about 0.06 during the first iterations and recovers once the network has fitted the data. With "linear", α becomes negative after the first step (≈ -0.39 , which is unphysical) and then recovers to the same estimate as "square". "square" is the only mapping that both keeps $\alpha > 0$ and avoids the collapse, which is why it is the default.

4 Loss

4.1 PDE residual

The residual is computed pointwise by automatic differentiation and vectorized over the collocation points with `jax.vmap`:

```
def pde_residual(layers, alpha, t, x, L=1.0):
    T_t = jax.grad(temperature, argnums=1)(layers, t, x, L)
    T_xx = jax.grad(jax.grad(temperature, argnums=2), argnums=2)(layers, t, x, L)
    return T_t - alpha * T_xx
```

The derivatives are exact, with no finite differences, and they are taken through the full ansatz, including the sin term and the prefactor.

4.2 Total loss

$$\mathcal{L}(\theta, \beta) = \underbrace{\frac{1}{N_f} \sum_{j=1}^{N_f} R(x_j, t_j)^2}_{\mathcal{L}_{\text{phys}}} + \lambda \underbrace{\frac{\sum_{i=1}^{N_d} w_i (T_\theta(x_i, t_i) - T_i^{\text{obs}})^2}{\sum_{i=1}^{N_d} w_i}}_{\mathcal{L}_{\text{data}}}, \quad (5)$$

with $\lambda = 1$. The weights default to $w_i = 1$, in which case $\mathcal{L}_{\text{data}}$ is the ordinary MSE. The forward problem uses $\mathcal{L}_{\text{phys}}$ only. Dividing by $\sum w_i$ keeps the scale of $\mathcal{L}_{\text{data}}$ independent of the weighting law.

The loss function returns the total loss and an auxiliary dictionary holding $\mathcal{L}_{\text{phys}}$ and the *unweighted* data MSE $\frac{1}{N_d} \sum_i (T_\theta - T_i^{\text{obs}})^2$. These are the loss curves shown in the figures. The unweighted MSE is used so that runs with and without weights can be compared on the same scale.

5 Optimization

Training uses `jaxopt.LBFGS` with the settings below. There is no Adam warm-up: L-BFGS starts directly from the initialized weights.

Setting	Value
<code>maxiter</code>	3000
<code>tol</code> (gradient-norm stopping criterion)	10^{-9}
<code>history_size</code>	10
<code>linesearch</code>	"zoom" (strong Wolfe conditions)
<code>Precision</code>	float64 (<code>jax.enable_x64=True</code>)

The loop is written explicitly so that the history can be recorded:

```
solver = LBFGS(fun=loss, has_aux=True, maxiter=3000, tol=1e-9,
               history_size=10, linesearch="zoom")
state = solver.init_state(params)
update = jax.jit(solver.update)
for it in range(maxiter):
    params, state = update(params, state)
    history["loss"].append(float(state.value))
    history["loss_phys"].append(float(state.aux["phys"]))
    history["loss_data"].append(float(state.aux["data"]))
```

```

history["alpha"].append(float(get_alpha(params, alpha_param=alpha_param)))
if not np.isfinite(history["loss"][-1]) or float(state.error) < tol:
    break

```

Training stops when the gradient norm falls below `tol`, when the loss becomes non-finite, or at `maxiter`. The forward run reported below used all 3000 iterations. There are no other early-stopping criteria.

Double precision matters: with float32, the second derivatives and the small late-time temperatures ($T \sim 10^{-2}$ to 10^{-3}) lose several significant digits.

6 Sensitivity-weighted inverse training

`inverse.fit_inverse_two_stage` works in two stages:

1. **Stage 1.** A standard inverse fit with $w_i = 1$ gives α_1 .
2. **Weights.** For each observation,

$$w_i = \max\left(\left(\frac{|S_\alpha(x_i, t_i; \alpha_1)|}{\max_k |S_\alpha(x_k, t_k; \alpha_1)|}\right)^p, w_{\min}\right), \quad p = \frac{1}{2}, \quad w_{\min} = 0.05. \quad (6)$$

The exponent $p = 1/2$ softens the concentration around the sensitivity peak; $w_{\min}$ keeps weakly informative points in the loss. The weights are then frozen.

3. **Stage 2.** A new fit from scratch, with the same initialization seed, collocation points and α_0 , using these weights in $\mathcal{L}_{\text{data}}$. The result α_2 is the reported estimate.

The weights enter the data term only. The PDE residual must hold everywhere, including in regions that carry little information about α . Weighting the IC/BC would be meaningless here anyway, since they are imposed exactly.

Important caveat. S_α is evaluated with its *analytical* formula, taken at the PINN estimate α_1 and never at the true α . In a real problem the analytical formula is unavailable. S_α would then have to be estimated from the PINN itself, for example by finite differences between PINNs trained at $\alpha_1 \pm \delta$, or by solving the sensitivity equation $\partial_t S = \alpha \partial_{xx} S + \partial_{xx} T$.

7 Evaluation and figures

- **Forward error.** The prediction is evaluated on a uniform 200×200 grid over $[0, t_{\max}] \times [0, L]$. The script reports the relative error $\|T_\theta - T\|_2 / \|T\|_2$ and the maximum absolute error.
- **Inverse error.** The script reports $|\hat{\alpha} - \alpha| / \alpha$, where $\hat{\alpha}$ is the value after the last iteration.
- **Animations.** A callback records $T_\theta(x, t_k)$ on 120 points in x . It does so at 60 log-spaced iterations (`np.geomspace(1, maxiter, 60)`), plus the first and last iterations. Log spacing is used because L-BFGS changes the solution mostly in the first tens of iterations. The GIFs play at 10 fps and hold the final frame for 20 frames.
 - *Forward GIF:* profiles at $t_k \in \{0, 0.15, 0.4, 0.8, 1.5\}$, next to the physics loss.
 - *Inverse GIF:* three panels. (i) Profiles at $t_k \in \{0, 0.4, 0.8\}$ of the sensitivity-weighted run, with the observations coloured by their time t_i . They are plotted against x only, so they do not lie exactly on the displayed profiles. (ii) α against iteration for the three variants of Section 8. (iii) Physics loss (solid) and unweighted data MSE (dashed) for the same three variants.

8 Results

8.1 Forward problem

For $\alpha = 0.18$ with seed 0 and 3000 iterations, the final physics loss is 1.38×10^{-7} . The relative L^2 error is 6.34×10^{-5} and the maximum absolute error 6.17×10^{-5} . The run takes 33s on a laptop CPU when nothing else is running.

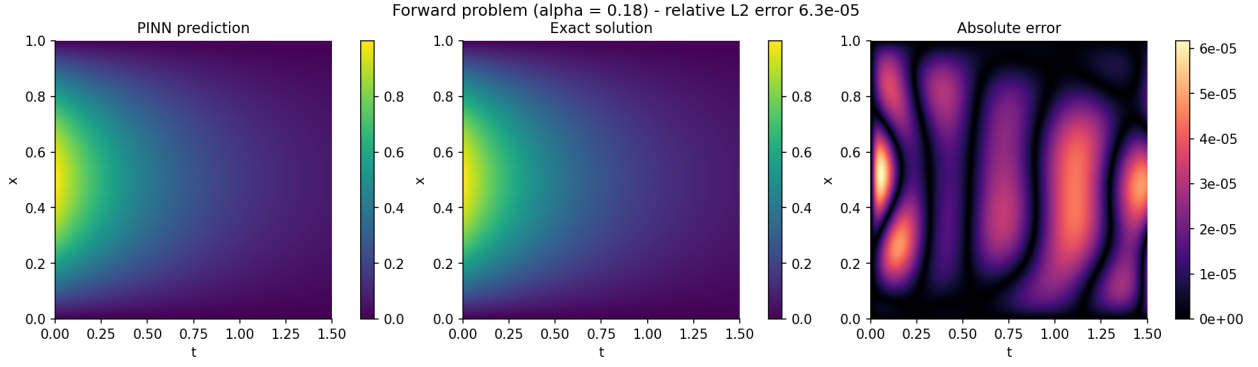


Figure 1: Forward problem, python scripts/forward.py.

8.2 Inverse problem, single dataset

scripts/inverse.py solves one dataset with three variants: "square" unweighted, "square" sensitivity-weighted (weights from the unweighted estimate), and "softplus" unweighted. Default settings: $\alpha = 0.5$, $\eta = 20\%$, data seed 0, 25 points drawn of which 18 are kept, $\alpha_0 = 1.0$.

Variant	$\hat{\alpha}$	relative error
"square", unweighted	0.44991	10.02%
"square", sensitivity-weighted	0.46899	6.20%
"softplus", unweighted	5.9×10^{-32}	100% (collapse)

This dataset was chosen for the figure because the weighting makes a visible difference; averages over many datasets are given in Section 8.3. For a more typical case ($\alpha = 0.18$, $\eta = 10\%$, 24 of 25 points kept), the unweighted and weighted fits give 0.17800 (1.11%) and 0.17881 (0.66%). With "softplus", the same collapse to $\alpha \approx 5 \times 10^{-32}$ occurs.

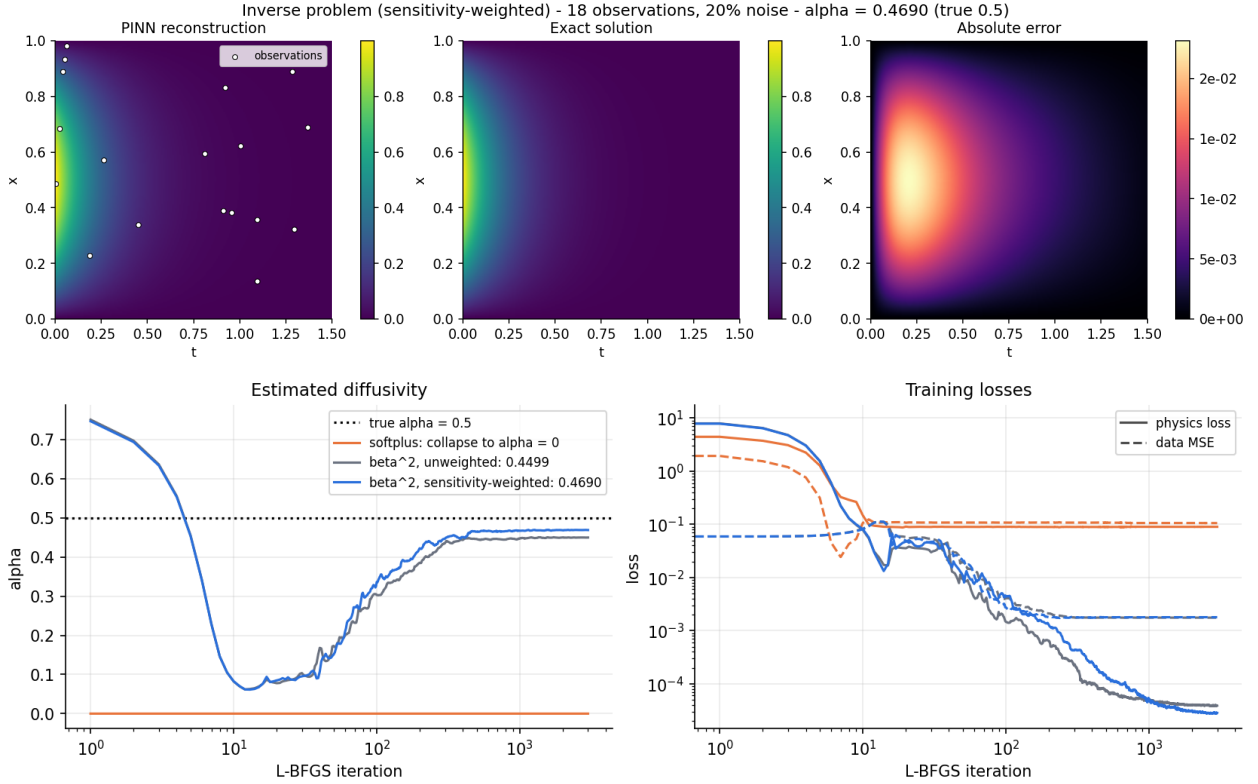


Figure 2: Top: sensitivity-weighted inverse reconstruction. Bottom: α and loss histories of the three variants (python scripts/inverse.py).

8.3 Robustness study

Protocol (`scripts/robustness_study.py`):

- $\alpha \in \{0.18, 0.5\}$, $\eta \in \{5, 10, 20\}\%$ and data seeds $\{0, 1, 2\}$, with $N_d = 25$ drawn points and $\alpha_0 = 1.0$.
- Each dataset is solved unweighted and with the two-stage weighting, which gives 36 results and 54 L-BFGS trainings in total.
- The network initialization seed and the collocation seed are both fixed at 0 for every run. The spread in the results therefore comes *only* from the data (sampling locations and noise realization), not from the optimization.

α_{true}	noise	unweighted	sensitivity-weighted
0.18	5%	0.47 (0.62)	0.49 (0.88)
0.18	10%	0.94 (1.23)	0.98 (1.76)
0.18	20%	2.31 (2.76)	2.26 (3.63)
0.5	5%	0.73 (1.01)	0.48 (0.59)
0.5	10%	2.35 (4.81)	1.56 (2.93)
0.5	20%	5.05 (10.02)	3.34 (6.20)

Table 1: Relative error on $\hat{\alpha}$ (%): mean (max) over 3 datasets of 25 points, $\alpha_0 = 1.0$.

Discussion. Three observations stand out.

- **Accuracy.** α is recovered within 0.5% at 5% noise and within a few percent at 20% noise, starting from a guess that is 2 to 6 times too large. The error grows with the noise level and varies noticeably between datasets.
- **Effect of the weighting.** It depends on the regime. For $\alpha = 0.18$ it changes nothing measurable (mean errors within 0.05 percentage points). For $\alpha = 0.5$ it reduces the mean error by about one third at every noise level, and the worst case at 20% noise from 10.0% to 6.2%. This matches the sensitivity analysis: faster diffusion shrinks the window $t \sim t^* = 1/(\alpha\pi^2)$ in which measurements are informative, so down-weighting uninformative points matters more.
- **Systematic bias.** In all 36 cases α is *underestimated*. A likely cause is the removal of negative measurements. At late times the true temperature is close to zero, so the discarded points are preferentially those with negative noise. The retained late-time data are therefore biased upward, which mimics slower decay, i.e. a smaller α . Testing this would require repeating the study without the filtering.

9 Reproducibility

Software	Version used
Python	3.11.14
JAX / jaxlib	0.9.2
jaxopt	0.8.5
NumPy	2.4.3
SciPy	1.17.0
pandas	3.0.0
Matplotlib	3.10.8

All runs were performed on CPU. Every source of randomness is seeded: the network initialization (JAX key), the collocation points (LHS seed) and the dataset (NumPy generator). Rerunning a script on the same software stack reproduces the numbers above. Small differences can appear with other versions or hardware because of floating-point reduction order.

10 Differences from the original research code

The repository is a condensed rewrite of a larger set of exploratory scripts. The method is the same: hard IC/BC ansatz, tanh network, Glorot initialization, float64, L-BFGS (`jaxopt`, zoom line search, history 10), LHS collocation, $\alpha = \alpha_{\min} + \beta^2$, removal of negative observations, and square-root sensitivity weighting. The deliberate differences are:

- **Sensitivity weights.** The original weighting study evaluated S_α at α_{fit} , a least-squares fit of the *exact solution* to the data. This rewrite evaluates it at the PINN’s own stage-1 estimate, which is closer to a realistic workflow. A minimum weight $w_{\min}$ is also added.
- **Data weight λ .** Here $\lambda = 1$. The original study used the empirical law $\lambda = 0.8(0.1/\eta)^3$, which depends on the noise level.
- **Random number generation.** Datasets use `np.random.default_rng` instead of `np.random.seed`, and network layers use split keys instead of the fixed per-layer seeds $100 + k$. The sampled points and initial weights therefore differ from those of the original study, and its numbers are not directly comparable.
- **Not included.** Early stopping based on MSE thresholds; soft-constraint and normalized formulations; Galerkin-type constraints; local noise estimation; the TensorFlow and PyTorch versions.

11 Known limitations

- The sensitivity used for weighting relies on the analytical expression of S_α (see above).
- λ , the architecture and the number of collocation points were not tuned in this rewrite.
- Robustness is assessed over datasets only; a single network and collocation seed is used.
- The results are for a linear 1D problem with a single-mode analytical solution. They do not by themselves transfer to nonlinear or multi-dimensional problems.
- `jaxopt` is no longer actively developed; the L-BFGS implementation in `optax` is a natural replacement.